CS & IT ENGINEERING



Operating System

CPU Scheduling

DPP 01 Discussion Notes



By- Saheb Tung sir

[NAT]

#Q. Consider the following process scenario.

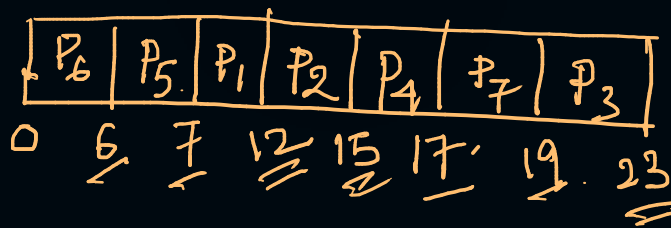
Process	Arrival Time (in milliseconds)	Burst Time (In milliseconds)	CT	TAT	WT
P1	4	5	12	8	3
P2	5	3	15	10	7
P3	8	4	23	15	11
P4	7	2	17	10	8
P5	3	1	7	4	3
P6	0	6	6	6	0
P7	7	2	19	12	10

Avg WT $\frac{72}{7} = 6$

The average waiting time of processes for FCFS scheduling algorithm is __6__ milliseconds?

Ans 6

Gantt. chart ⇒ FCFS. ⇒ non preemptive.



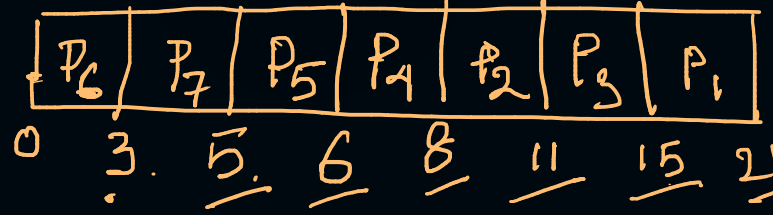
[NAT]

#Q. Consider the following process scenario.

Process	Arrival Time (in milliseconds)	Burst Time (In milliseconds)	CT.	TAT.	WT.
P1	5	6	21	16	10
P2	3	3	11	8	5
P3	1	4	15	14	10
P4	2	2	8	6	4
P5	4	1	6	2	1
P6	0	3	3	3	0
P7	1	2	5	4	2
					32

The average waiting time processes for non-preemptive shortest job first scheduling algorithm is 4.57 milliseconds (rounded up to 2 decimal places)?

Gantt chart → SJF ⇒ non-preemptive.



Avg WT = 32/7

= 4.571 ms

[NAT]

#Q. Consider a CPU performance metric throughput which is calculated as:

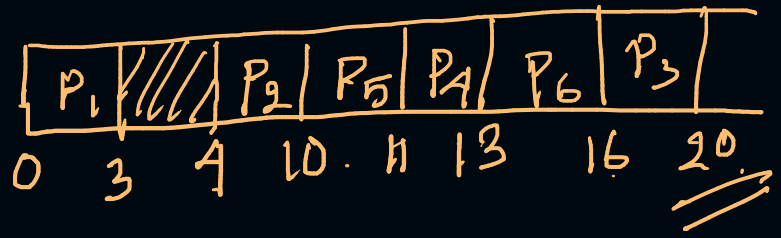
$$\text{Throughput} = \frac{\text{Number of processes executed}}{\text{Total scheduling duration from first process arrival till last process completion}}$$

For the following process scenario calculate the throughput calculated for non-preemptive SJF algorithm 0.3 per milliseconds (rounded up to 1 decimal place)?

6 process.

Process	Arrival Time (in milliseconds)	Burst Time (in milliseconds)
P1	0	3
P2	4	6
P3	7	4
P4	9	2
P5	8	1
P6	6	3

Gantt-chart of SJF



Throughput = $\frac{6}{20}$

$= 0.3$

[NAT]

#Q. Consider a CPU performance metric throughput which is calculated as:

$$\text{Throughput} = \frac{\text{Number of processes executed}}{\text{Total scheduling duration from first process arrival till last process completion}}$$

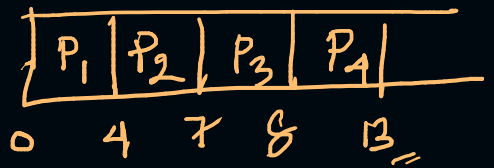
For the following process scenario calculate the throughput calculated for FCFS algorithm as x and for non-preemptive SJF scheduling as y. The value of x – y is ____?

Process	Arrival Time	Burst Time
P1	0	4
P2	0	3
P3	0	1
P4	0	5

Ans 0.

FCFS.

Gantt chart.

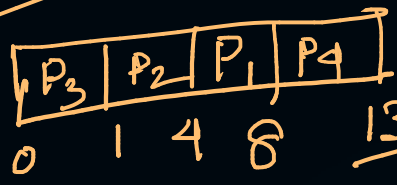


Throughput = $\frac{4}{13}$.

$x = \frac{4}{13}$

SJF

Gantt chart.



Throughput = $\frac{4}{13}$

$y = \frac{4}{13}$

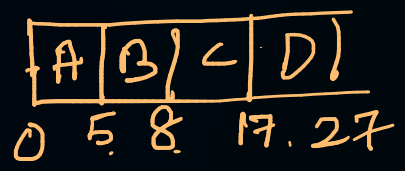
$x - y = 0$

[NAT]

#Q. Consider 4 processes A, B, C and D. All arrived at time 0 in the given order. The processes needed 5ns, 3ns, 9ns and 10ns respectively for their CPU burst to complete. The average turnaround time of processes if executed in FCFS order is 14.25 ns (rounded up to 2 decimal place)?

	AT	BT	CT	TAT
A	0	5	5	5
B	0	3	8	8
C	0	9	17	17
D	0	10	27	27
				57

Gantt chart ⇒ FCFS.



$$\text{Avg TAT} = \frac{57}{4} = 14.25$$

[NAT]

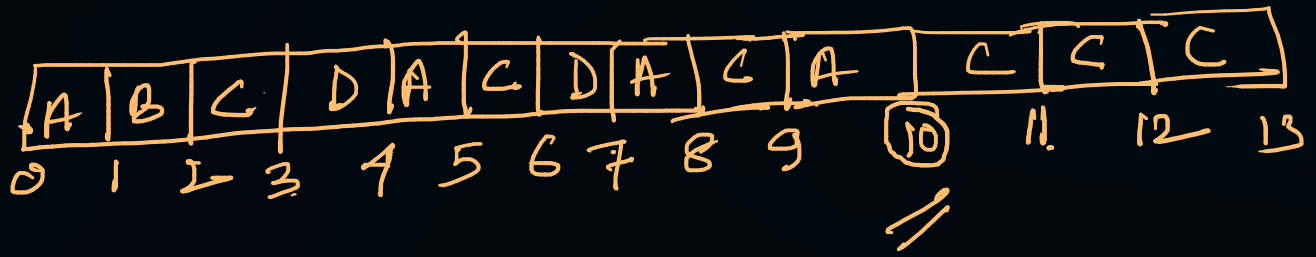
#Q. Four processes to be executed on a single processor system arrives at time 0+ in the order A, B, C and D. Their CPU burst time requirements are 4, 1, 6, 2 time units respectively. The completion time of ~~processe~~ A under Round-Robin scheduling with time slice of one time unit is 10 ?

t = 1 time unit.

	AT	BT
A	0+	4 3 2 1
B	0+	1
C	0+	6 5 4 3 2 1
D	0+	2 1

Ready queue

t=0 | A B C D



Ans 10

[MCQ]

#Q. On a system using round robin CPU scheduling, context-switch overhead is given by 's'. Time quantum is 'q'. The CPU efficiency, if q=s is?

A. 50%

B. Zero

C. 100%

D. Not predictable

CPU efficiency = $\frac{q}{s+q} = \frac{q}{q+q} = \frac{q}{2q} = .5 = 50\%$

Ans A

[NAT]

#Q. Consider the following process scenario.

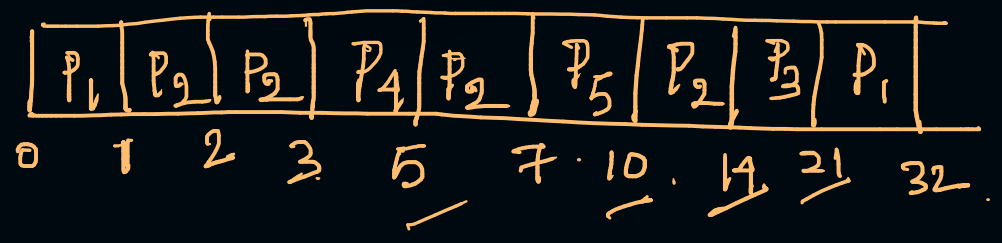
Process	Arrival Time (in milliseconds)	Burst Time (In milliseconds)	CT	TAT	WT
P1	0	12	32	32	20
P2	1	8	14	13	5
P3	2	7	21	19	12
P4	3	2	5	2	0
P5	7	3	10	3	0
					37

The average waiting time processes for preemptive shortest remaining time first scheduling algorithm is 7.4 milliseconds (rounded up to 1 decimal place)?

Ans 7.4

Avg WT = 37/5 = 7.4

Gantt chart → SRTF → preemptive.



[NAT]

#Q. Consider the following process scenario.

Process	Arrival Time (in milliseconds)	Burst Time (In milliseconds)	CT	TAT	WT
P1	0	8	16	16	8
P2	1	4	19	13	9
P3	2	1	7	5	4
P4	4	5	18	14	9
					30

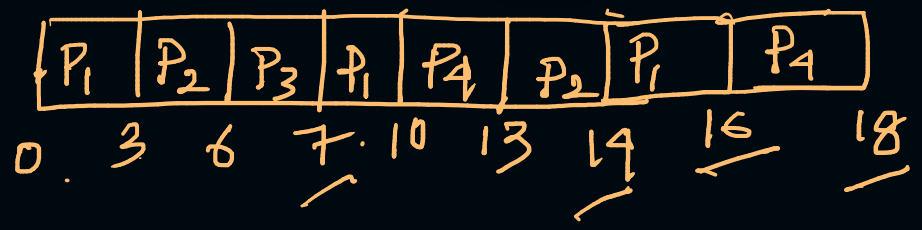
The average waiting time processes for round robin scheduling algorithm is 7.5 milliseconds with time slice of 3 milliseconds (rounded up to 1 decimal place)?

Ans 7.5

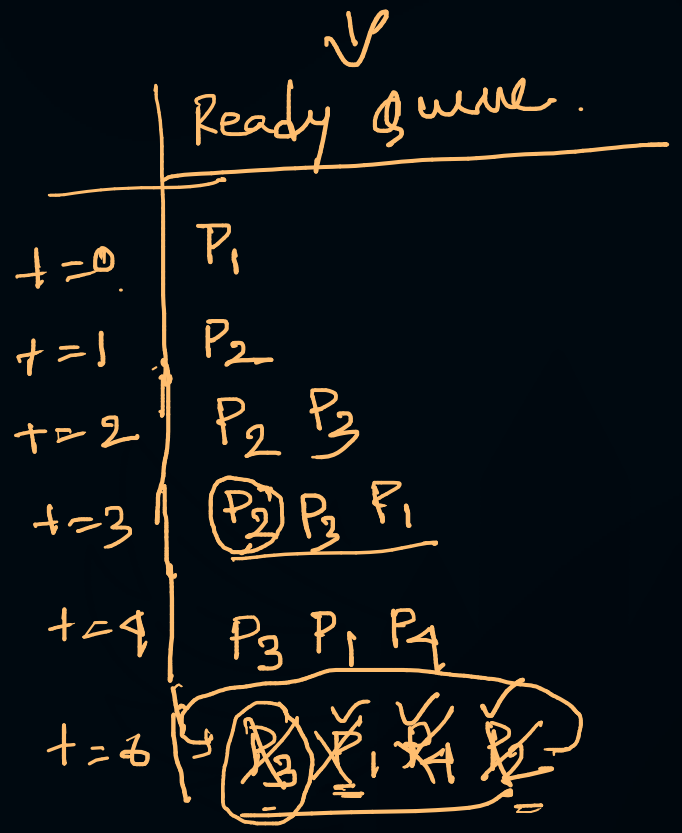
Avg WT = $\frac{30}{4}$ = 7.5

$t = 3$.

Gantt-chart ⇒ RR ⇒ preemptive.



	BT
P_1	~~8~~ ~~5~~ ~~2~~
P_2	~~4~~ ~~1~~
P_3	1
P_4	~~5~~ ~~2~~



[NAT]

#Q. A computer system has 2GB of RAM and OS occupies 256MB of RAM. All the user processes are of 128MB and have same characteristics. If the goal is 99% CPU utilization, then the maximum I/O wait that can be tolerated is 72 % (rounded to nearest integer)?

CPU utilization $= 1 - p^n$

$n = 14$

$.99 = 1 - p^{14}$

$p^{14} = .01$

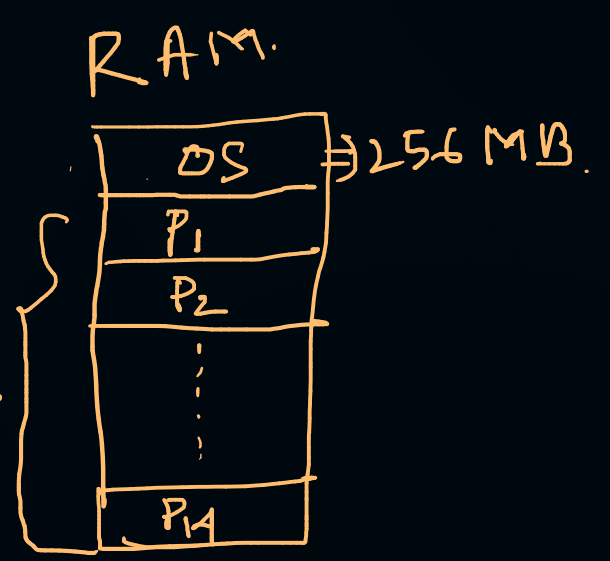
$p = (.01)^{1/14}$

$= .7196$

$= 71.96\%$

$= 72\%$

Ans 72



Total no of process $= 7 \times 2 = 14$

$= \frac{256\,MB}{128\,MB} = 2$

$2GB = 2^{31}\,B$

$\frac{2^{31}}{256MB} = \frac{2^{31}\,B}{2^{28}\,B} = 2^3 = 8$

THANK - YOU